Venkata Rohan Kambhampati

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Portfolio | LinkedIn | GitHub

Professional Summary

AI Engineer with 4 years building production LLM, RAG, and multi-agent systems on Microsoft Azure and Google Cloud. Lead development of a **Google ADK** agent on Gemini (Vertex AI) that executes **7K+ enterprise actions per month** through **MCP** servers and **A2A** on GKE, with verified-identity controls, human-in-the-loop approvals, and PII-safe logging. Previously delivered **70% RAG latency reduction**, **55% LLM cost savings**, and **95% intent-routing accuracy** for 1,000+ enterprise users.

Experience

Equinix (Feb 2023 – Present)

Staff AI Engineer | *Google ADK, Gemini (Vertex AI), A2A, MCP, Python, GKE, Helm* Mar 2026 – Present

- **Led Development of DIO Agent:** Built a **Google ADK** multi-agent system on Gemini 3.5 Flash that routes IT, procurement, and finance requests to 4 sub-agents (ServiceNow, Coupa, Oracle ERP, SAP Fieldglass) over 29 MCP tools; exposed via A2A and deployed on GKE, executing **7K+ enterprise actions per month** on an agentic platform with **9.5K+ monthly users**; extended ADK callbacks so 24 card-returning tools skip a redundant LLM pass.
- **Engineered Enterprise MCP Servers:** Primary author of the **Coupa** MCP server (11 approval, lookup, receipt, and soft-close tools) handling **2.4K+ purchase-request approvals per month**, re-validating record-level authorization before every write and requiring **human-in-the-loop** confirmation; built an Oracle ERP vendor-statement tool with fail-closed org-unit access control.
- **Hardened Agent Identity & PII-Safe Logging:** Fixed attribution spoofing in Coupa writes using verified session identity; ported signed-session verification to the Oracle and Fieldglass MCP servers; built PII/secret redaction (JWTs, SSNs, Luhn-validated card numbers) and correlated JSON logging across 5 services.
- **Shipped Agents to Production:** Rolled out the agent and 4 MCP servers via **Helm** GitOps with Workload Identity; diagnosed a dev outage of the I/O-bound agent and added memory + CPU HPA autoscaling with tolerant liveness probes.

Senior Associate Software Engineer | *Python, FastAPI,Azure OpenAI, OpenAI Agents SDK, Azure AI Search* Jun 2023 – Mar 2026

- **Production RAG (Equinix GPT):** Built hybrid retrieval with semantic reranking on **Azure AI Search** over 15K+ ServiceNow, Confluence, and SharePoint documents (44K chunks); cut latency from 8.5s to **2.5s** via parallel retrieval and server-side vectorization; synced ServiceNow updates with Azure Functions.
- **RAG Evaluation:** Built a retrieval eval harness (181-query stratified golden set, MRR, nDCG, Hit@k with bootstrap CIs); reached **0.90 Hit@5** and **0.74 MRR@10**, +56% over BM25, with an LLM judge confirming relevant context in the top 5 for 95% of queries; audited all 44K chunks, surfacing 4.2K duplicates, 175 stale chunks, and 76% embedded without article titles, each mapped to a fix.
- **LLM Cost Optimization:** Reduced monthly OpenAI API spend by **55%** ($12K → $5.4K) through prompt compression and output-token limits.
- **ERP Workflow Automation:** Built Node.js microservices integrating SAP Fieldglass, Coupa, and Microsoft Teams Adaptive Card approvals; automated 15K+ monthly supplier-onboarding tasks, cutting approval time from **5 days to 6 hours**.

Software Developer Intern | *React, React Flow, Oracle Integration Cloud* Feb 2023 – Jun 2023

- Built a React Flow DAG visualizer that cut deployment errors by 30%; automated ledger-posting reconciliation via Oracle Integration Cloud, saving 24 hours of manual effort per month.

Projects

Distributed Stock Exchange Simulation (GitHub) | *TypeScript, Node.js, RabbitMQ, PostgreSQL, Docker*

- Engineered a 4-microservice trading platform (API, order-book engine, DB service, WebSocket server) with RabbitMQ queues decoupling matching from persistence and WebSocket pub-sub for real-time price updates.

Open Source Contributions

Lightdash (PR #12119): Alert links open the threshold editor directly, cutting configuration steps by 30%.
Cal.com (PR #18192): Preserved custom booking-question order on the booking confirmation page.
Rocket.Chat (PR #32475): Built "Notify on Reactions" alerts for reactions to a user's messages.

Technical Skills

Agentic AI & LLM: Google ADK, A2A Protocol, Model Context Protocol (MCP), FastMCP, OpenAI Agents SDK, Gemini (Vertex AI), Azure OpenAI, Prompt Engineering, Guardrails, Human-in-the-Loop
RAG & Evaluation: Vector & Hybrid Search, Azure AI Search, Semantic Reranking, Retrieval Metrics (MRR, nDCG, Hit@k), LLM-as-Judge
Cloud & DevOps: Google Cloud (GKE, Vertex AI, Workload Identity), Azure (Functions, App Service, Service Bus), Kubernetes, Helm, GitOps, Docker, GitHub Actions
Security: OAuth 2.0 (Token Exchange, Introspection), PingFederate, Service Identity, PII Redaction
Backend & Data: FastAPI, Node.js, Express.js, PostgreSQL, Redis, MongoDB, Azure Cosmos DB, RabbitMQ, Microservices, Event-Driven Architecture
Languages & Frontend: Python, TypeScript, JavaScript, Java, SQL, React.js, Next.js

Education

B.Tech in Computer Science and Engineering, Mahindra University, Hyderabad CGPA: 8.5 | July 2019 – May 2023

Achievements

Solved **1,100+** competitive programming problems (LeetCode, GFG, InterviewBit) | Deep Learning Specialization (Coursera)